Natnael Kornima

Full-Stack Engineer & Systems Builder

Addis Ababa, Ethiopia · Open to Remote & Relocation

Email: natnaelkornima78@gmail.com
GitHub: github.com/natnaelkornima
LinkedIn: linkedin.com/in/natnaelkornima
Portfolio: natnaelkornima.github.io

PROFESSIONAL SUMMARY

Full-Stack Software Engineer specializing in scalable web architectures, resilient distributed backends, and deterministic AI/LLM system integrations. Extensive experience with TypeScript, Next.js, Node.js, Python, PostgreSQL, and event-driven queues. Focused on building high-throughput services with sub-100ms latency, rigorous type safety, and clean user interfaces.

CORE TECHNICAL EXPERTISE

Languages: TypeScript, JavaScript (ESNext), Python, SQL, HTML5/CSS3

Frontend: React, Next.js, Tailwind CSS, Responsive Design, Web Audio API

Backend: Node.js, Fastify, Express, Python (FastAPI), BullMQ

Databases & Queues: PostgreSQL, Redis, Qdrant Vector DB, Prisma/ORM

AI & Data Systems: LangChain, Vector Embeddings (RAG), Pydantic Guardrails

DevOps & Tools: Docker, Git/GitHub, Linux/Bash, REST APIs, CI/CD

SELECTED ENGINEERING PROJECTS

Nazazi SMS & School Management Platform · Full-Stack — Next.js · Node.js · Fastify · PostgreSQL · Redis · BullMQ

- Architected a decoupled, event-driven notification queue with BullMQ and Redis handling burst SMS traffic and automated retries.
- Implemented idempotent dispatch tracking in PostgreSQL, guaranteeing zero duplicate dispatches and sub-100ms ingestion latency.

Enzira Begena E-Learning Platform · Full-Stack & Audio — TypeScript · React · Next.js · Node.js · PostgreSQL · Web Audio

- Built a customized indigenous music education platform featuring Web Audio API interactive notation, audio synthesis, and lesson sequences.
- Normalized relational curriculum schema in PostgreSQL and optimized multimedia buffering for seamless low-bandwidth playback.

Neural RAG & Vector Guardrails · AI & Backend — Python · FastAPI · Qdrant · Redis · LangChain · Pydantic

- Built an asynchronous RAG pipeline combining BM25 keyword matching with dense vector embeddings and cross-encoder reranking.
- Enforced strict runtime output parsing with Pydantic guardrails and Redis semantic caching, reducing inference latency by 60%.

NutriGenius AI Dietary Intelligence · AI & Full-Stack — Python · FastAPI · LangChain · NumPy · Tailwind CSS

- Engineered an NLP parsing engine translating natural meal descriptions into structured JSON schemas calibrated against nutritional tables.

EXPERIENCE & WORK HISTORY

Full-Stack Software Engineer — Independent Systems & Client Applications — 2024 — Present | Addis Ababa / Remote

- Designed and delivered distributed web architectures, high-deliverability messaging gateways, and specialized e-learning platforms.
- Engineered containerized deployment workflows using Docker, automated unit/integration testing suites, and relational migrations.

Software Developer — Technical Project Development — 2023 — 2024 | Addis Ababa

- Developed modular REST microservices with Node.js and Express, implementing cryptographic JWT authentication and access control.
- Constructed responsive, accessible client interfaces conforming to WCAG 2.1 AA accessibility standards.

EDUCATION & CREDENTIALS

Computer Science & Software Engineering Focus — Higher Education Ongoing / Completed

Coursework: Data Structures, Algorithms, Database Systems, Computer Networks, Operating Systems, Distributed Architecture.